

ISE 5970: Advanced Systems Optimization

University of Oklahoma School of Industrial and Systems Engineering

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Instructions for Assignment Submission:

1. The assignment must be uploaded to Canvas by the indicated due date. The solution should be uploaded to Canvas as a PDF file.
2. Along with your main submission (in PDF format), please attach any supplementary files (Excel, Python) that were instrumental in your work. If any Python code was used, upload it in HTML format.

Assignment 3: Decomposition Techniques and Formulating General Optimization Problems

1. **(25 points) (Extra credit) Dantzing-Wolfe Decomposition** A transnational soda company meets a demand of $1,000 \text{ m}^3$ by operating three local factories in different countries. The soda is produced by blending four key ingredients: the company's trademark formula, mineral water, fruit juice, and brown sugar. While the trademark formula is supplied centrally at a fixed cost of \$8 per m^3 , the other three ingredients are sourced locally at country-specific costs.

The cost structure for the local ingredients is summarized in the following table:

Table 1: Cost of Ingredients in Each Country (Excluding Trademark Formula)

Ingredient	Country 1	Country 2	Country 3
Mineral Water	\$2.70	\$2.50	\$3.10
Fruit Juice	\$3.00	\$3.60	\$3.10
Brown Sugar	\$3.20	\$4.00	\$4.10

Each m^3 of soda produced must adhere to the following composition requirements:

- **Fruit Juice:** between 25% and 35%
- **Brown Sugar:** between 10% and 30%
- **Trademark Formula:** between 2% and 4%

Due to its complex production process, the trademark formula is available in a maximum quantity of 25 m^3 .

Furthermore, each country imposes its own regulatory limits on the soda's content of hydrocarbons and vitamins. The content is determined by the sum of the products of each ingredient's volume and its corresponding coefficient. The coefficients for hydrocarbons and vitamins for each ingredient are provided below:

Table 2: Coefficients for Hydrocarbons and Vitamins per Ingredient

	Trademark Formula	Mineral Water	Fruit Juice	Brown Sugar
Hydrocarbons	0.10	0.07	0.08	0.09
Vitamins	0.10	0.05	0.07	0.08

The regulatory limits on the total hydrocarbon and vitamin content in the soda for each country are as follows:

Table 3: Hydrocarbon and Vitamin Limits per Country

	Country 1	Country 2	Country 3
Max Hydrocarbons	24	27.5	30
Min Vitamins	19.5	22	22

- (a) **(5 points)** Formulate the linear programming (LP) model for the soda production problem.
- (b) **(5 points)** Explain why this problem is suitable for a Dantzig-Wolfe decomposition, and describe the decomposition structure (i.e., the division into subproblems and the master problem).
- (c) **(15 points)** Solve the problem to optimality using the Dantzig-Wolfe decomposition approach. Provide the results and detail the process for each iteration.
2. **(25 points) Benders Decomposition** Consider a two-stage stochastic procurement problem for a factory that must secure enough energy to meet its demand for both the current year and the following year. The factory uses two types of fuel—coal and natural gas—measured in MWh. The decision process is as follows:
- First-Stage Decisions (Year 1)** At the beginning of the first year, the factory procures coal and natural gas to satisfy a known demand of 850 MWh. The current prices are:

- Coal: \$4.5 per MWh
- Natural Gas: \$5.1 per MWh

Second-Stage Decisions (Year 2) The energy demand for the second year is uncertain and will be revealed at the beginning of that year. Upon realization, a corrective procurement decision is made. The possible outcomes are:

- **High Demand:** 950 MWh (probability 0.3)
Prices: Coal at \$7.5 per MWh, Natural Gas at \$8.5 per MWh
- **Medium Demand:** 780 MWh (probability 0.4)
Prices: Coal at \$6 per MWh, Natural Gas at \$5.5 per MWh
- **Low Demand:** 530 MWh (probability 0.3)
Prices: Coal at \$3 per MWh, Natural Gas at \$4 per MWh

All prices are expressed in terms of energy production per MWh.

Balancing Constraint To ensure a balanced energy mix, the total amount of coal or natural gas (procured over both years) must be between one-third and two-thirds of the total energy demand (i.e., the sum of the demands for both years).

- (a) **(5 points)** Formulate the linear programming (LP) model for the coal and natural gas.
- (b) **(5 points)** Explain why this problem is suitable for Bender's decomposition, and describe the decomposition structure (i.e., the division into subproblems and the master problem).
- (c) **(15 points)** Solve the problem optimally using Bender's decomposition approach. Provide the results and detail the process for each iteration.
3. **(25 points) Formulation Problem** Consider a routing problem defined on a set of 10 cities labeled $A, B, C, D, E, F, G, H, I$, and J . Two planes are available for covering these cities. The routing decisions must satisfy the following conditions:
- **Starting Cities:**
 - Plane 1 starts and ends its route at city A .
 - Plane 2 starts and ends its route at city F .
 - Each city (except the starting cities) must be visited exactly once by one of the two planes.
 - The goal is to design the routes for both planes to minimize the total travel distance.
 - Plane 1 has a limited fuel capacity, which constrains the total flight distance to be less than 500 units.

The distances between the cities are given by the symmetric distance matrix below:

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>
<i>A</i>	0	12	8	15	10	20	18	16	14	22
<i>B</i>	12	0	11	9	14	21	19	17	13	24
<i>C</i>	8	11	0	13	9	19	16	15	12	20
<i>D</i>	15	9	13	0	10	17	14	12	16	23
<i>E</i>	10	14	9	10	0	18	15	13	11	19
<i>F</i>	20	21	19	17	18	0	8	6	10	12
<i>G</i>	18	19	16	14	15	8	0	7	9	11
<i>H</i>	16	17	15	12	13	6	7	0	8	10
<i>I</i>	14	13	12	16	11	10	9	8	0	14
<i>J</i>	22	24	20	23	19	12	11	10	14	0

- (a) **(15 points)** Formulate the integer programming (IP) model for the routing assignment. Include sets, decision variables, parameters, objective functions, and constraints.
- (b) **(10 points)** Solve the formulated model using Gurobi and report the plane travel schedule and the objective function value.
4. **(25 points) Column generation** You work for a large finance firm and are tasked with building a portfolio for a key client. The portfolio must include a specified number of shares from eight major US-based companies. Because of your firm's size, you can purchase shares at preferential rates but with a twist: the bank sells the shares only in packages. Each package is valued at up to \$1,000 and can contain a mix of shares from different companies, as long as the total cost does not exceed \$1,000. Finally, regardless of how the package was built, the bank is going to charge you \$900 each.

Your objectives are twofold:

- **Package Design:** Determine the composition of each package you want to buy (that is, the number of shares of each company per package) so that the total cost is within the limit of \$1,000.
- **Portfolio Assembly:** Decide how many of each type of packages to purchase to meet the general demand for shares while minimizing the total cost.

The following table summarizes the target demands and the cost per share for each company.

Company	Demand (Shares)	Cost per Share (USD)
Apple Inc.	30	110
Microsoft Corp.	20	150
Amazon.com Inc.	50	160
Alphabet Inc.	40	240
Meta Platforms Inc.	35	120
Tesla Inc.	25	220
JPMorgan Chase	60	80
Berkshire Hathaway	15	350

Table 4: Share Demands and Costs for the Portfolio

For your solution, assume that you can buy fractional units of a package but to build each package use an integer number of shares.

- (a) **(5 points)** Formulate the integer programming (IP) model for the portfolio problem.
- (b) **(15 points)** Define a feasible initial base and solve the problem using the column generation technique. Explicitly explain the reformulation required, and at each iteration show the packages used as well as the solution.
- (c) **(5 points)** Imagine now that you can also build packages with fractions of shares. Resolve the problem again and report your findings.

5. **(25 points) Formulation: Mosaic Builder** A computer image is made up of hundreds of pixels or colored dots that, when combined, create the impression of continuous color. The color of each pixel is commonly encoded in the RGB scale, where each color can be represented as an ordered triplet. Each component is an integer between 0 and 255 and represents the saturation of red, green, and blue in that order, so the triplet (0,0,0) represents black or absence of color, and the triplet (255,255,255) represents white.

Using a simple program that allows extracting information from an image (.BMP, .JPG) into a comma-separated values table (and viceversa), such that

- all pixels are arranged on a Cartesian plane with the origin at the top-left corner of the image, as shown in Figure 2, and
- a table is generated showing the X coordinate of the pixel, the Y coordinate of the pixel, the saturation value of the red color, the saturation value of the green color, and the saturation value of the blue color,

you must solve the following problems.

- (a) (10 points) Formulate a linear mathematical program that allows you to convert an image with $X_1 \times Y_1$ pixel resolution to another with $X_2 \times Y_2$ resolution. Figure 2 shows an example of a 5:1 scale reduction. Use some of the images available in Canvas and at least one image of your choice for testing.

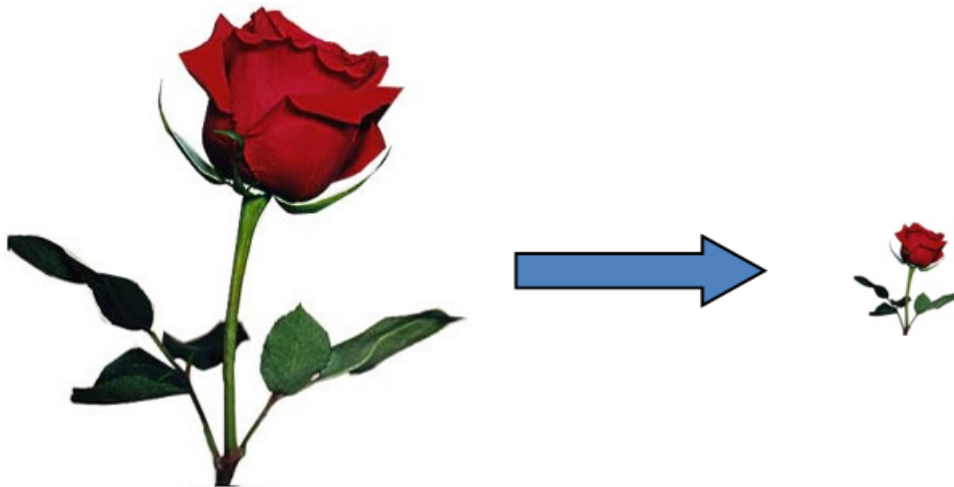


Figure 1: Example of “optimal” photo resizing

- (b) (10 points) Formulate a linear program that allows you to convert a color image into an image created solely from other images (known as a photo mosaic effect). To create this photo mosaic image, you must use a database with at least 10 different images. Additionally, the resulting image (photo mosaic) must be generated with at least 100 images in total. For instance, the photo mosaic could be a 10×10 image grid but may include repeated images. Figure 2 shows two examples of the photo mosaic effect. Use some of the images available in Canvas and at least one image of your choice for testing.

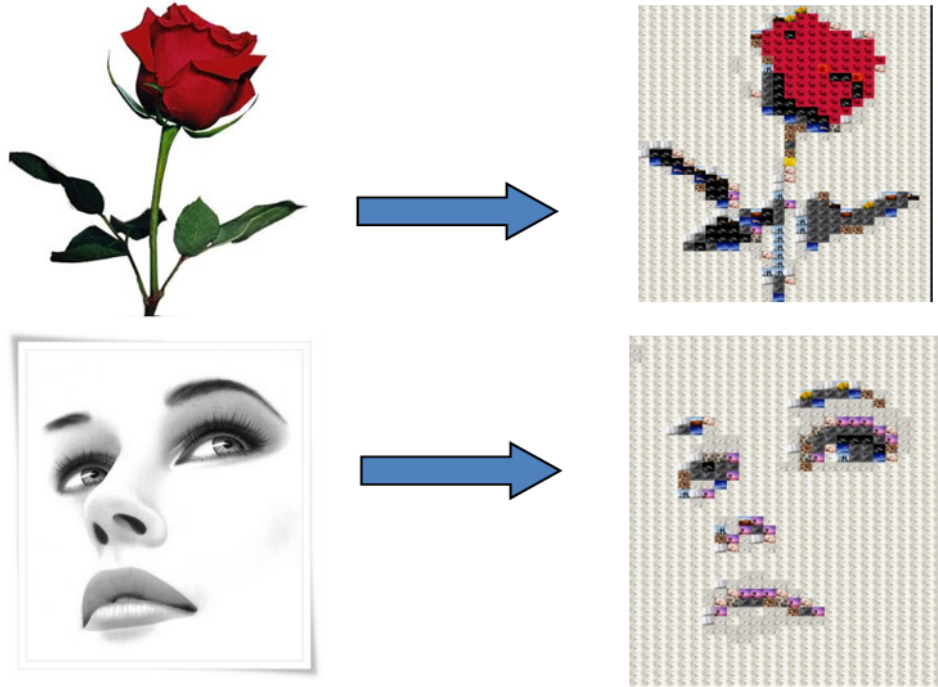


Figure 2: Examples of “optimal” photo mosaic

- (c) (5 points) Formulate a linear program that allows you to perform a sensitivity analysis (constructing a Pareto efficiency frontier) of the photo mosaic effect (using the objective function from the previous problem) as a function of the number of images used. Show the Pareto front constructed for at least two images used in the previous problem.

For each problem, you must show your formulation (sets, parameters, decision variables, objective function, constraints), implement your formulation in Gurobi/Python, attach your outputs to the appendices, and present the resulting images in your written report.